

Math 300 Exam 1a

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (9 pts) Express each sentence in sentential logic using proper notation:

a) Either A or both B and C.

b) A exactly when B.

c) Exactly one of A or B.

2. (6 pts) Consider the five basic truth functions $f_{\sim}$, $f_{\wedge}$, $f_{\vee}$, $f_{\rightarrow}$, and $f_{\leftrightarrow}$, to compute the value of each composite function below. Show some steps. Do not just write down a value.

a) $f_{\rightarrow}(f_{\vee}(T, F), f_{\wedge}(T, F)) =$

b) $f_{\sim}(f_{\leftrightarrow}(T, f_{\vee}(F, T)) =$

3. (5 pts) Prove the argument $p \vee q, p \therefore q$ is invalid.

4. (18 pts) Answer the following questions about standard truth tables.

a) How many lines will there be in the standard truth table with 4 variables?

b) What is a sentence called if it has truth value T for every line of the table?

c) What is a sentence called if it has truth value F for every line of the table?

d) What is a sentence called if it has truth value T for some lines and F for other lines?

e) What does it mean if two sentences, A and B, are logically equivalent?

f) What is the maximum number of statements with 3 variables such that no two of the statements are logically equivalent?

5. (4 pts) Find a formal sentence using only the connectives $\sim$ and $\wedge$ that is logically equivalent to the sentence $\sim(p \rightarrow q)$

$$\sim(p \rightarrow q) \equiv$$

6. (4 pts) Construct a formal sentence B that satisfies the following truth table.

p	q	r	B
<i>T</i>	<i>T</i>	<i>T</i>	<i>T</i>
<i>T</i>	<i>T</i>	<i>F</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>F</i>	<i>F</i>
<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
<i>F</i>	<i>T</i>	<i>F</i>	<i>T</i>
<i>F</i>	<i>F</i>	<i>T</i>	<i>F</i>
<i>F</i>	<i>F</i>	<i>F</i>	<i>T</i>

B =

7. (8 pts) Rewrite each statement and its contrapositive **as an if-then statement**.

a) Centre wins when they score at least 70 points.

Rewrite:

Contrapositive:

b) Every differentiable function is continuous.

Rewrite:

Contrapositive:

8. (8 pts) Construct a five-step proof to show $A \rightarrow (\sim C), B \wedge C \therefore \sim A$ is a valid argument. Label each step using the proper name for the rules of deduction used.

1.

2.

3.

4.

5.

9. (10 pts) Prove the statement: If $a \equiv b \pmod{n}$ and x is any integer, then $ax \equiv bx \pmod{n}$.

10. (6 pts) Consider the set $\{a, b, c, d, e, f, g, h, i, j\}$ with subsets $A = \{a, e, i\}$, $B = \{b, c, d, e, g\}$, and $C = \{b, d, f, j\}$. List the elements in each of the following sets. Use proper notation.

a) $A \times A =$

b) $C \setminus B =$

c) $B^c \cup A =$

11. (10 pts) Prove or disprove the statement: If x^2 is an irrational number then x is also an irrational number.

12. (8 pts) Write both of the theorems below **as if-then sentences**.

a) The zero product property:

b) The division algorithm:

13. (4 pts) List all the numbers between -10 and 20 that are congruent to 5 mod 7